



Spatial-temporal three-dimensional ultrasound plane-by-plane active cavitation mapping for high-intensity focused ultrasound in free field and pulsatile flow



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ABSTRACT

Cavitation plays important roles in almost all high-intensity focused ultrasound (HIFU) applications. However, current two-dimensional (2D) cavitation mapping could only provide cavitation activity in one plane. This study proposed a three-dimensional (3D) ultrasound plane-by-plane active cavitation mapping (3D-UPACM) for HIFU in free field and pulsatile flow. The acquisition of channel-domain raw radio-frequency (RF) data in 3D space was performed by sequential plane-by-plane 2D ultrafast active cavitation mapping. Between two adjacent unit locations, there was a waiting time to make cavitation nuclei distribution of the liquid back to the original state. The 3D cavitation map equivalent to the one detected at one time and over the entire volume could be reconstructed by Marching Cube algorithm. Minimum variance (MV) adaptive beamforming was combined with coherence factor (CF) weighting (MVCF) or compressive sensing (CS) method (MVCS) to process the raw RF data for improved beamforming or more rapid data processing. The feasibility of 3D-UPACM was demonstrated in tap-water and a phantom vessel with pulsatile flow. The time interval between temporal evolutions of cavitation bubble cloud could be several microseconds. MVCF beamformer had a signal-to-noise ratio (SNR) at 14.17 dB higher, lateral and axial resolution at 2.88 times and 1.88 times, respectively, which were compared with those of B-mode active cavitation mapping. MVCS beamformer had only 14.94% time penalty of that of MVCF beamformer. This 3D-UPACM technique employs the linear array of a current ultrasound diagnosis system rather than a 2D array transducer to decrease the cost of the instrument. Moreover, although the application is limited by the requirement for a gassy fluid medium or a constant supply of new cavitation nuclei that allows replenishment of nuclei between HIFU exposures, this technique may exhibit a useful tool in 3D cavitation mapping for HIFU with high speed, precision and resolution, especially in a laboratory environment where more careful analysis may be required under controlled conditions.

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1. Introduction

High-intensity focused ultrasound (HIFU) has emerged as a noninvasive technique to selectively and locally produce therapeutic effects in a specific site of the body without affecting the intervening tissue layers [1,2]. It has a promising potential in its applications, such as tumor ablation [3], extracorporeal lithotripsy [4], hemostasis [5], thrombolysis [6] and tissue erosion [7]. As cavitation plays important roles in both of thermal [8,9] and mechanical effects [10,11] which are the two principal mechanisms of

HIFU therapy, a better understanding of the spatial-temporal distribution of cavitation bubbles is essential to optimize these applications.

For decades, techniques for detecting and mapping cavitation have been developed, and these techniques are mainly divided into optical and acoustic methods [12,13]. Optical methods, such as high-speed photography (HSP), sonoluminescence (SL) and sonochemiluminescence (SCL), have been employed to image the cavitation field. HSP can capture images of bubble dynamics and distribution when ultrasound exposure is on, and this powerful tool has a sufficiently high frame rate [14,15]. Compared to high-speed photography, SL and SCL can provide more information about the spatial distribution of chemically “active” cavitation

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bubbles [16,17]. In addition to its high cost, HSP, SL and SCL, however, are limited in transparent media, which triggers for seeking alternative methods of cavitation detection.

Acoustic methods take advantage of the acoustical characteristics of bubble activities, such as the sub-harmonic and broadband noise corresponding to the stable and the inertial cavitation, and thus are more suitable for *in situ* studies [18,19]. Passive cavitation detection (PCD) and active cavitation detection (ACD) techniques are two common one-dimensional acoustic detection methods with high sensitivity and spatial specificity [20]. However, they cannot reflect the spatial distribution of cavitation. Recently, PCD and ACD have been adapted to use in cavitation mapping techniques, through employing ultrasound imaging arrays as the cavitation detectors, specifically in passive cavitation mapping (PCM) and active cavitation mapping (ACM). PCM generally uses diagnostic ultrasound scanners with a linear array which receives only to increase spatial coverage [21–24]. The passive images of cavitation field during HIFU exposures can be reconstructed by beamforming the array element signals and time-exposure acoustics (TEA) methods [23–25]. Several groups have approached PCM by performing passive beamforming in the time domain. The cavitation features can also be characterized by transforming cavitation signals from the time domain to the Fourier domain [22]. PCM can be used to monitor cavitation activity during the sonication, providing information about the mode and strength of the oscillations. Arvanitis et al. used a transcranial MRI-guided focused ultrasound (MRgFUS) system to visualize stable and inertial cavitation in the brain transcranially [26,27]. However, in the transcranial cases, PCM cannot reconstruct the bubble cloud without making some correction for field aberrations imposed by the overlying bone.

ACM approaches mainly refer to conventional diagnostic B-mode ultrasound imaging which enables detection and localization of bubble activity [28]. However, strictly simultaneous ACM cannot be achieved by the conventional B-mode imaging, because of its sequential scanning process. Additionally, since the imaging frame rate is limited, conventional B-mode imaging cannot capture the transient behavior of cavitation bubbles. Gateau et al. have proposed an ultrafast active cavitation mapping (UACM) based on plane wave transmission and delay-and-sum (DAS) beamforming, and they combined it with passive detection to detect and locate the cavitation both *in vitro* and *in vivo* [29,30]. A tilted plane wave compounding method was also proposed to improve the signal-to-noise ratio (SNR) and resolution of the bubble images [29,31]. As reported in another study, Hu et al. have developed an UACM with relatively high spatial-temporal resolution, through combining plane wave transmission, minimum variance (MV) beamforming and coherence factor (CF) weighting to monitor the evolution of residual cavitation bubbles in histotripsy [32]. However, as the ultrasound beams are unfocused, lateral resolution of the ultrafast images needs to be improved. In order to develop a standard cavitation mapping similar to the exact acoustic field measurement, we have presented a modified line-by-line scanning ACM to map cavitation bubbles with both high sensitivity and spatial-temporal resolution [33]. Li et al. have proposed a new ACM termed bubble Doppler, which is based on a fusion of the adaptations of three Doppler techniques, i.e., color Doppler, pulse-inversion Doppler, and interleaving Doppler. This technique can both spatially map the presence of transient bubbles and estimate their sizes and the degree of nonlinearity for pulsed HIFU (pHIFU) applications [34].

However, two-dimensional (2D) cavitation mapping could only provide cavitation activity in one plane. In fact, cavitation bubbles are generated in a three-dimensional (3D) volume, which means that the 3D cavitation map is necessary to provide more information of cavitation activity. O'Reilly et al. provided 3D transcranial PCM using a sparse hemispherical array, which was capable of

monitoring bubble emissions down to single bubble events through an *ex vivo* human skull [35]. In addition, using noninvasive phase correction and super-resolution image processing, transcranial PCM of brain vascular can be achieved [36]. Collin et al. presented a near-acoustically-transparent, 2D 32-element PVDF array suitable for use as a 3D cavitation mapping sensor, which can be mounted on the therapy transducer of a clinical HIFU device [37]. However, the number of elements and received data for construction of 3D image are greatly increased, which would increase the cost of the instrument and thus restrict the application.

In this study, we proposed a 3D ultrasound plane-by-plane active cavitation mapping (3D-UPACM) based on the linear array of a current ultrasound diagnosis system. The acquisition of channel-domain raw radio-frequency (RF) data in 3D space is performed by sequential plane-by-plane 2D ultrafast active cavitation mapping. Between two adjacent unit locations, there is a waiting time to allow the cavitation nuclei to distribute within the liquid and it can return to its original state. With this spatial series of channel-domain raw RF data, a 3D cavitation map can be reconstructed that encompasses the entire cavitation volume to be detected at one time. MV adaptive beamforming, is combined with CF weighting (MVCF) or compressive sensing (CS) method (MVCS) to process the raw RF data for improved beamforming or more rapid data processing. This 3D-UPACM technique is demonstrated in tap-water and a phantom vessel with the pulsatile flow. Temporal evolutions and spatial distributions of cavitation bubble cloud have been observed. The performance of 3D-UPACM is evaluated in the 2D-UPACM from resolution, SNR level and time penalty. This 3D-UPACM allows 3D visualization of the time evolution of bubble activity, using the linear array of a current ultrasound diagnosis system rather than a 2D array transducer, therefore decreasing the cost of the instrument. Although the application is limited by the requirement for a gassy fluid medium or a constant supply of new cavitation nuclei that allows replenishment of nuclei between HIFU exposures, this technique may exhibit a useful tool in 3D cavitation mapping for HIFU with high speed, precision and resolution, especially in a laboratory environment where more careful analysis may be required under controlled conditions.

2. Materials and methods

2.1. 3D ultrasound plane-by-plane active cavitation mapping

In the 3D-UPACM method, cavitation event is mapped immediately after one HIFU exposure by a linear array of the ultrasound imaging system in plane wave transmitting and receiving mode, and then extended this by sequentially measuring a series of planes. These planes consist of the linear array positioned at different unit locations perpendicular to the axis of the HIFU transducer. Between two adjacent unit locations, there is a waiting time to make cavitation nuclei distribution of the liquid back to the original state. After acquiring the spatial series of channel domain raw radio frequency (RF) data, we can obtain a 3D cavitation map equivalent to the one detected at one time and over the entire volume. The acquisition of the raw RF signals for each unit location is synchronized with HIFU exposure. The duration of HIFU exposure, as well as the delay of the interrogating pulse relative to the moment while HIFU is turned off, can vary from microseconds to seconds.

The 3D-UPACM that we undertake would strictly only be valid when the cavitation distribution is repeatable if all conditions including HIFU output and the physical properties of surrounding medium are same. It has been proved that ultrasound, host media, and nuclei parameters are important in determining cavitation activity *in vitro* [38–40]. In addition, cavitation is highly complex

with transient characteristics, and cavitation bubbles may generate, grow, merge and collapse. Also cavitation bubble can migrate and accumulate due to acoustic radiation forces, and such accumulation may in turn affect the incident sound field which goes on to alter the form of the subsequently cavitation [41]. Given these properties of cavitation, the cavitation cloud is unlikely to be perfectly reproduced. However on the broad scale of the mapping done here, it may be valid to build up a 3D cavitation map plane-by-plane, and this is here tested by re-measuring for specific test cases in tap-water. The tests may be useful to show to what extent the cloud is repeatable over the timescales that we are assuming it is. The experiments of the specific test cases were repeatedly operated for 20 times. The time interval between repeats was about 5 min and no insonification occurred between repeats.

2.2. Experimental set-up

A schematic of the experimental setup was shown in Fig. 1. It consisted of the HIFU exposure system and ultrasound imaging system with plane wave. Experiments were performed in a tank ($60 \times 40 \times 30$ cm) made of Lucite. Acoustic absorbers (RTV silicone rubber, provided by Institute of Acoustics, Chinese Academy of Sciences) were placed on the walls of the water tank to minimize the potential effects of reflections on acoustic field. All the experiments were operated at the room temperature (20 ± 2 °C).

2.2.1. HIFU exposure system

The HIFU system consisted of a focused single element concave transducer (Imasonic, Besancon, France), a power amplifier

(AG1017, T&G Power Conversion, Inc., Rochester, NY) and a double channel arbitrary waveform generator (AWG420, Tektronix). Sinusoidal signal generated by one channel of the double channel arbitrary waveform generator was amplified by the power amplifier. The electric output drove the focused transducer to generate cavitation. The transducer mounted on one side of the tank had a curvature of 120-mm radius, a geometric focal length of 97.4 mm, and a working frequency of 1.2 MHz at which the transmit efficiency (acoustical power/electrical load power) was 72% (provided by the manufacturer). The acoustic power was estimated from electrical load which was displayed in the panel of the power amplifier. The acoustic field distribution of HIFU transducer was measured in the degassed water tank using a needle hydrophone (0.2-mm diameter, Precision Acoustics Ltd., Dorchester, UK) with the electrical power set to 2 W. Acoustic pressure distribution measured was described in detail elsewhere [42]. The focused ultrasound field could be divided into the pre-focal region, the focal region and the post-focal region. The full-width half-maximum (FWHM) beam length and width of the transducer are 8 mm and 1.6 mm respectively. At high electrical power, the acoustic pressure cannot be measured directly by a needle hydrophone for fear of damaging it. Meanwhile, the generation of cavitation under high power may make it hardly feasible to characterize the pressure field because of the strong nonlinear characteristic and the erosion of violent cavitation activities. Therefore, the *in situ* spatially averaged intensity (defined as I_{SAL} by Hill et al.) is estimated using the following equation [43]:

$$I_{SAL} = 0.867 \frac{W}{D^2} \quad (1)$$

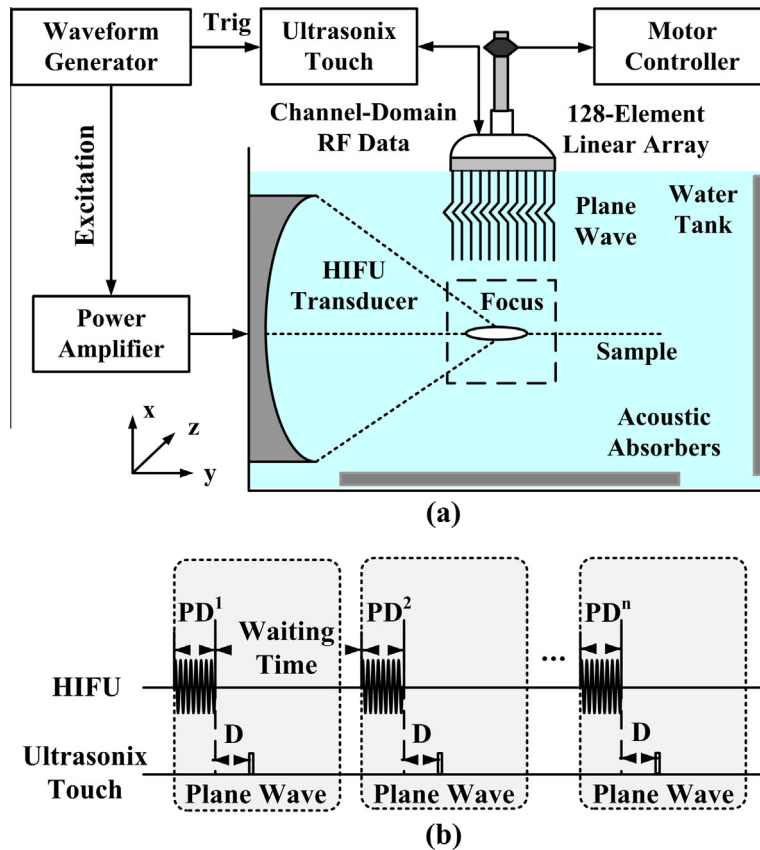


Fig. 1. Schematic diagram of the experimental setup for plane-by-plane 3D cavitation mapping. (a) Experimental setup. (b) Time-sequences of the synchronization of HIFU exposure, RF data acquisition of Ultrasonix-Touch. In (b), PD and D were the duration of HIFU exposure and the delay of the Ultrasonix-Touch's triggering pulse relative to the moment while HIFU was turned off.

where W is the total acoustic power output of the transducer and D is the half pressure maximum beam width. Acoustic powers and the corresponding spatial average intensities at different electrical power levels could be estimated according to Eq. (1) and here, a constant acoustic power of 72 W, corresponding to the calculated spatial average intensity of 2438 W/cm^2 at therapeutic levels, is used to perform the HIFU exposure, and large enough to generate intense cavitation bubble cloud.

2.2.2. Ultrasound imaging system with plane wave

A SonixTouch ultrasound scanner (Ultrasonix Medical Corp., Richmond, BC, Canada), equipped with a 128-element linear transducer array (L14-5/38 probe, 38 mm aperture, center frequency of 7.5 MHz, 65% bandwidth) and a plug-in module for parallel data acquisition of the raw ultrasound data (SonixDAQ, Ultrasonix Medical Corp., Richmond, BC, Canada), was used to map the cavitation bubble population immediately after one HIFU exposure by programmable plane wave transmission (5 MHz, 1 cycle) at around 90 kPa, which was sufficiently weak not to yield acoustic driven diffusion. Meanwhile, it was strong enough to maintain a reasonable SNR. The time delay between cavitation excitation and ultrasound plane wave transmission was 1 ms, and the sampling rate of the array was 40 MHz. Using a three-axis stepper motor controller (Zolix SC300-3B, Beijing, China), the 128-element linear array can be positioned at each unit location perpendicular or parallel to the axis of HIFU transducer. Here, the distance between two adjacent locations was set to 0.5 mm.

2.2.3. Timing synchronization

As shown in Fig. 1(a), sinusoidal signal generated by one channel of the double channel arbitrary waveform generator was amplified by the power amplifier to drive the HIFU transducer. Another channel of the generator was used to trigger raw RF data acquisition of SonixDAQ. Therefore, the acquisition of the raw RF signals for each unit location was synchronized with HIFU exposure. Timing-sequences of the synchronization of HIFU exposure and RF data acquisition of SonixDAQ were illustrated in Fig. 1(b). PD and D were respectively, the duration of HIFU exposure, as well as the delay of the SonixDAQ triggering pulse relative to the moment while HIFU was turned off. Through hardware control, PD and D could vary from microseconds to seconds. Images of the cavitation bubble cloud at its different evolution stages and the dissolving process of corresponding bubbles were acquired through varying the parameters PD and D respectively. In order to simplify the experimental operation, there was a sequence of HIFU exposures with different PDs at each unit location. Between two successive HIFU exposures of different PDs, there was a waiting time long enough to make cavitation nuclei distribution of the liquid back to the original state.

2.3. Sample preparations

This 3D-UPACM was demonstrated both in tap-water and a phantom vessel with the pulsatile flow.

2.3.1. Fresh tap water

The tank was filled with fresh tap water with the temperature varied within $19.1\text{--}20.2^\circ\text{C}$ and Oxygen content of the water varied within $4.6\text{--}5.0 \text{ mg/L}$ during the whole experiment. The objective of this sample was to investigate the 3D distribution of HIFU-induced cavitation under free-field conditions. During 3D-UPACM proceeding, the 128-element linear array was successively positioned at each unit location perpendicular to the axis of HIFU transducer and 40 locations were used for 3D acquisition.

2.3.2. Transparent wall-less flow phantom

A polyacrylamide wall-less phantom was employed to mimic the flow environment and soft tissue surrounding the vessel. The ingredients and proportions required to create a 200 ml gel were listed in Table 1. Since thermal effect was not concerned in this study, no protein was added into the phantom for HIFU. The measured density of the phantom was $1.003 \pm 0.002 \text{ g/cm}^3$, and the ultrasound speed and attenuation coefficient at 1 MHz in phantom were $1513 \pm 2 \text{ m/s}$ and $0.20 \pm 0.01 \text{ dB/cm}$, respectively. The wall-less flow phantom was made by the following steps. First, a cylinder rod was inserted into a hollow box to create a vascular channel. Second, the hollow box was filled with the gel mixture liquid. After the mixture liquid is solidified, the cylinder rod was removed from the box carefully to produce a cylindrical, wall-less flow vessel with internal diameters of the rod size. In this study, the diameter of the vessel was 4 mm. The 0.9% saline solution was pushed into the vessel by a pulsating pump (Model 1405, Harvard Apparatus, USA) with an average flow velocity range of $0\text{--}4.24 \text{ cm/s}$. The phantom was then mounted in an acrylic holder connected to a high resolution X–Y–Z positioning stage (D120, Suruga Seiki Co., Ltd., Tokyo, Japan) with the vessel aligned perpendicularly to the acoustic axis. When the experiment was performed in the phantom vessel with pulsatile flow, the Lucite tank was filled with degassed water to reduce the influence of non-degassed water on the HIFU field through cavitation.

The HIFU focus was positioned in the middle of the vessel as the following steps. First, a 0.1 mm T-type thermocouple was inserted into the phantom with its tip located exactly in the front of the vessel. During the HIFU sonication at acoustic power of 2 W, the maximal temperature was observed by moving the phantom with the X–Y–Z positioning stage, which meant that the ultrasound focus was correctly positioned at the thermocouple tip. Second, the thermocouple was removed, and the location of the phantom was controlled by the X–Y–Z positioning stage. After the HIFU focus was positioned in the middle of the vessel, the linear array was aligned parallel to the vessel. With the three-axis stepper motor controller, the linear array can be positioned at each unit location parallel to the axis of HIFU transducer and 20 locations were used for 3D acquisition.

2.4. Data processing

MV adaptive beamforming is combined with CF weighting or CS method to process the raw RF data for improved beamforming or more rapid data processing. The schematic representation of data processing is shown in Fig. 2. A comb notch filter is employed to filter out the fundamental and harmonic HIFU frequencies in order to eliminate the strong interference from the HIFU field. To improve both the resolution and SNR of cavitation map, adaptive beamforming, including time delay, MV adaptive beamforming and CF weighting, are used to process the raw RF data. With aim to retain the quality of adaptive beamforming and reduce the calculating data involved, MV beamforming processing is accomplished on random resampling points from a certain bandwidth, rather than on every sampling point. And then the total

Table 1
Chemical composition of 200 ml polyacrylamide gel [44].

	Volume (ml)	Percent (%)
Degassed, distilled water	143.22	71.61
40% Acrylamide	35.00	17.50
1 M TRIS	20.00	10.00
10% APS	1.68	0.84
TEMED	0.10	0.05
Total volume	200	100

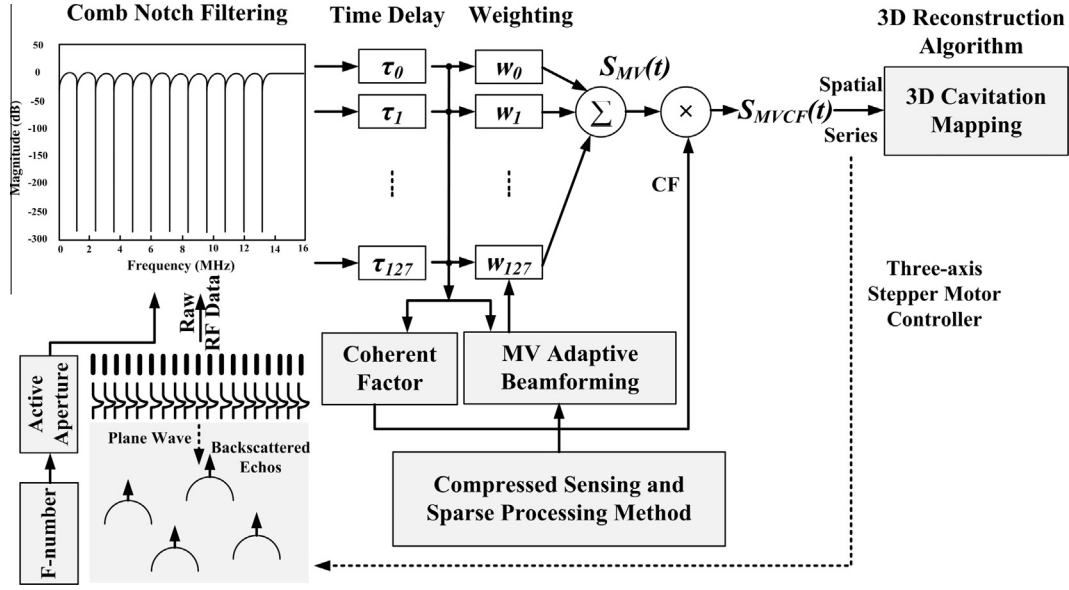


Fig. 2. Schematic data processing of 3D-UPACM.

information in frequency domain is reconstructed by the CS theory, which effectively reduced the volume of data involved in the beamforming calculation and made a significant increase in rendering speed. After acquiring each output of beamformed RF data in 3D space, the 3D cavitation map can be reconstructed by the 3D reconstruction algorithm.

2.4.1. Beamforming based on combined MV adaptive beamforming and CF weighting

The raw RF data acquired at a very high frame rate by plane wave transmission are beamformed to generate cavitation images. Due to the low energy of plane wave transmission and lack of transmit focusing, the traditional DAS beamforming performs poor resolution and low SNR. Therefore, MV adaptive beamforming and CF weighting are introduced in our work to improve the imaging quality [32].

The time delays are firstly calculated as the propagation time from the transmitting element to the focus point and back to the receiving element to achieve dynamic receiving focusing. Assuming an active aperture of M elements for a given point $\vec{r}_p$, the m th delayed element signal can be given as:

$$rf_m(t) = ch_m((\|\vec{r}^{xmt} - \vec{r}_p\| + \|\vec{r}^{rcv} - \vec{r}_p\|)/c), \quad (2)$$

for $m = 0, 1, \dots, M-1$, where $ch_m(t)$ is the received channel-domain raw RF signal on the m th element, c is the speed of sound, $\vec{r}^{xmt}$ and $\vec{r}^{rcv}$ are the spatial positions of the transmitting and receiving elements, respectively. The delayed element signals are weighted and summed to form beamformed RF data as follows:

$$S_{out}(t) = \sum_{m=0}^{M-1} w_m(t) rf_m(t) = \mathbf{w}(t)^H \mathbf{RF}(t), \quad (3)$$

where $w_m(t)$ is the weighting factor for element m , $\mathbf{w}(t) = [w_0(t) w_1(t) \dots w_{M-1}(t)]^T$, $\mathbf{RF}(t) = [rf_0(t) rf_1(t) \dots rf_{M-1}(t)]^T$, $()^H$ and $()^T$ denote the conjugate transpose and transpose, respectively.

Then, MV beamforming [45] is performed by computing the element weight adaptively by minimizing the variance of the beamformed output while maintaining a distortion-less response of the focal point. This optimum weight is given by:

$$\mathbf{w}_{opt}(t) = \frac{\mathbf{R}(t)^{-1} \mathbf{a}}{\mathbf{a}^H \mathbf{R}(t)^{-1} \mathbf{a}}, \quad (4)$$

where $\mathbf{R}(t)$ is the covariance matrix of $\mathbf{RF}(t)$, and $\mathbf{a}$ is the steering vector, which is simply a vector of ones due to time delay.

Spatial smoothing is used to de-correlate the coherent received signals, and diagonal loading is performed to improve the robustness of MV beamforming [46]. With spatial smoothing, the covariance matrix is estimated as:

$$\hat{\mathbf{R}}(t) = \frac{1}{M-L+1} \sum_{l=1}^{M-L+1} \mathbf{RF}_l(t) \mathbf{RF}_l(t)^H, \quad (5)$$

where L is the sub-array length, and $\mathbf{RF}_l(t) = [rf_l(t) rf_{l+1}(t) \dots rf_{l+L-1}(t)]^T$ is the delayed element vector for the l th sub-array. With diagonal loading, $\hat{\mathbf{R}}(t)$ is replaced by $\hat{\mathbf{R}}(t) + \varepsilon \mathbf{I}$, where ε is set to Δ -times of the power in the received signals $\varepsilon = \Delta \cdot \text{trace}\{\hat{\mathbf{R}}(t)\}$, $\mathbf{I}$ is a diagonal identity matrix.

The optimum weight can be obtained from Eq. (4) after substituting $\hat{\mathbf{R}}(t) + \varepsilon \mathbf{I}$ with $\mathbf{R}(t)$, and the MV beamformed output can be given as:

$$S_{MV}(t) = \frac{1}{M-L+1} \sum_{l=1}^{M-L+1} \mathbf{w}_{opt}(t)^H \mathbf{RF}_l(t). \quad (6)$$

Finally, to further enhance the resolution and SNR of cavitation image, CF weighting [47], another adaptive side-lobe-reduction method, is carried out through weighting the image pixel by the corresponding coherence factor:

$$S_{MVCF}(t) = CF(t) \cdot S_{MV}(t) = \frac{|\sum_{m=1}^M \mathbf{RF}(t)|^2}{M \sum_{m=1}^M |\mathbf{RF}(t)|^2} \cdot S_{MV}(t) \quad (7)$$

The beamformed RF data are then enveloped and scaled to form UACM images.

Although the imaging resolution and SNR can be improved by MV adaptive beamforming and CF weighting, the reconstruction time of it is tens of seconds or even hundreds of seconds in 2D imaging, and is dozens of times more in 3D imaging. Therefore, CS and sparse processing method is proposed to MV adaptive beamforming in our 3D cavitation mapping to boost the reconstruction speed.

2.4.2. CS and sparse processing method

Regular adaptive beamforming continuously updates the spatial filter for every sampling point, which refers to the huge amount of calculation and background processing normally be required, especially in the case with high sampling rate. Recently, the emerging CS technique indicates that sparse signals can be accurately reconstructed from a small amount of coefficients [48]. Here, MV beamforming combined with CS theory is used to retain the quality of adaptive beamforming and reduce the calculating data involved.

The useful frequency information concentrated in an available bandwidth rather than the entire frequency domain, which enlightened us to select sampling points in this available bandwidth to perform the beamforming processing. For further reduction in calculation, MV adaptive beamforming was accomplished on 40% points extracted randomly from the available bandwidth, representing the beamforming was carried on in a compressive domain. The beamformed data for UACI in the frequency domain was subsequently reconstructed by the CS technique and transformed into the time domain waiting for regular imaging procedure [49].

More specifically, according to the center frequency and its relative bandwidth, the available bandwidth K was selected first. And $\mathbf{RF}(\omega_k)$, $k = 1, 2, \dots, K$ was the corresponding covariance matrix. Then, we designed a diagonal matrix $\mathbf{T}$ by $K \times K$ as the extraction matrix, in which ones in the leading diagonal line were corresponding to extracting rows, and others were zeros. This can be demonstrated as follows:

$$\mathbf{RF}^H(\omega_{ex}) = \mathbf{T} \mathbf{g} \mathbf{RF}^H(\omega_k) \quad (8)$$

And the results of spatial smoothing and diagonal loading were:

$$\hat{\mathbf{R}}(\omega_{ex}) = \frac{1}{M-L+1} \sum_{l=1}^{M-L+1} \mathbf{RF}_l(\omega_{ex}) \mathbf{RF}_l^H(\omega_{ex}) \quad (9)$$

$$\hat{\mathbf{R}}(\omega_{ex}) = \hat{\mathbf{R}}(\omega_{ex}) + \varepsilon \mathbf{I} \quad (10)$$

Subsequently, the optimum weighting and the output on selected points were given by:

$$\mathbf{w}_{ex} = \frac{\hat{\mathbf{R}}^{-1}(\omega_{ex}) \mathbf{a}(\theta_0)}{\mathbf{a}^H(\theta_0) \hat{\mathbf{R}}^{-1}(\omega_{ex}) \mathbf{a}(\theta_0)} \quad (11)$$

$$\mathbf{y}(\omega_{ex}) = \frac{1}{M-L+1} \sum_{l=1}^{M-L+1} \mathbf{w}(\omega_{ex}) \mathbf{RF}_l(\omega_{ex}) \quad (12)$$

Moreover, according to the CS technique, the output $\mathbf{y}(\omega_{ex})$ can be expressed as [46]:

$$\mathbf{y}^H(\omega_{ex}) = \mathbf{T} \mathbf{y}^H(\omega_k) = \mathbf{T} \mathbf{F} \mathbf{b}^H(\omega_k) = \mathbf{\Phi} \mathbf{\Psi} \mathbf{b}^H(\omega_k) = \mathbf{\Theta} \mathbf{b}^H(\omega_k) \quad (13)$$

where $\mathbf{F}$ ($K \times K$) was the Fourier basis, $\mathbf{\Psi} = \mathbf{I}$ was the orthogonal basis, $\mathbf{\Phi} = \mathbf{T} \mathbf{F}$ was the measurement matrix and $\mathbf{\Theta} = \mathbf{\Phi} \mathbf{\Psi}$ was the sensing matrix.

Here, we utilized the Regularized Multiple Focal Underdetermined System Solver (R-M-FOCUSS) to estimate $\mathbf{b}(\omega_k)$ in Eq. (13) [50]. Finally the output $\mathbf{y}(\omega_k)$ for the available bandwidth was acquired, and was filled with zeros to obtain the entire frequency spectrum $\mathbf{y}(\omega)$. After transforming $\mathbf{y}(\omega)$ into time domain, the plane wave MVCS-based ACM with fine resolutions, SNR and rapid calculating can be accomplished as regular imaging procedure.

2.4.3. 3D reconstruction algorithm

As the spatial series of beamformed RF data in 3D space are acquired, the 3D cavitation map can be reconstructed by the 3D reconstruction algorithm. Marching Cube algorithm, can be used to visualize the cavitation bubbles in the 3D dataset [51]. As a computer graphics algorithm for extracting a polygonal mesh of

an isosurface from a 3D discrete scalar field, Marching Cube has been widely used in medical image visualization.

With a bunch of equidistant parallel scanned cavitation maps, we construct a 3D structured point dataset, and eight scalar values from two adjacent slices of our dataset are used to form the vertices of an imaginary cube. Here, the basic procedure of the Marching Cube algorithm for 3D cavitation mapping is as follows. First, an iso-value for the algorithm is set according to cavitation bubbles boundary pixel value we would like to extract. Second, for each imaginary cube in dataset we compare the scalar's value with the iso-value. If the scalar's value is higher than or equal to the iso-value, then we can say it falls within or on the surface. Otherwise, it falls outside the surface. After all eight scalars are checked, we can create a patch to represent the isosurface in this cube. Third, by connecting the patches from all cubes on the isosurface boundary, the surface representation for the cavitation bubbles can be obtained. Using the Visualization Toolkit (VTK) which is an open-source, freely available software system for 3D computer graphics, image processing and visualization, the 3D visualization of the cavitation bubbles can be obtained.

The impacts on image quality and reconstruction time of these several post processing steps of the proposed algorithm are demonstrated from both simulation and experimental data. The simulation is to use the lateral variations of the point targets (the point spread functions) to clearly demonstrate the impact of these steps in the strong scattering imaging. Also, one of the experimental images is chosen to demonstrate the impact of these steps in visualizing cavitation bubbles.

3. Results

3.1. The repeatability of the ultrasound plane-by-plane scanning method

Fig. 3 showed the results of re-measuring on three different timescales in tap-water at the location of 0 mm. The exposure time of HIFU in Fig. 3(a)–(c) were 50 μ s, 5 ms and 30 ms, respectively. In (a) and (b), the first row showed the RF signal of the 64th scan line for both the initial and re-measuring experiments. In (c), the first row showed the RF signal of the 120th scan line for both the initial and re-measuring experiments. As the results shown in the first row, the initial results for the 64th (Fig. 3(a) and (b)) and 120th (Fig. 3(c)) scan line could be almost reproduced at the end. Also, the second, third and fourth rows illustrated three times of cavitation maps obtained by ultrasound plane-by-plane scanning method. Because the single bubble event may randomly occur in all the area where acoustic pressure exceeded the cavitation threshold, cavitation clouds are unlikely to be totally identical. Nevertheless, these images appeared very similar and the bubble cloud structure did not change in quality. The bubble cloud is structurally stable on time-scales large compared to the bubble oscillations.

Furthermore, gray value curves along the linear array in tap-water at the location of 0 mm were plotted in Fig. 4. The values shown were the average of 20 measures, and the error bars showed the standard deviation. As shown in Fig. 4, the bubble cloud structure did not change in quality. Both the results shown in Figs. 3 and 4 may prove the repeatability of the method on the broad scale of the cavitation mapping done here.

3.2. Performance of 3D-UPACM

The performance of 3D-UPACM is mainly evaluated in the 2D-UPACM from resolution, SNR level and time penalty. The time penalty refers to the computing processing time of beamforming

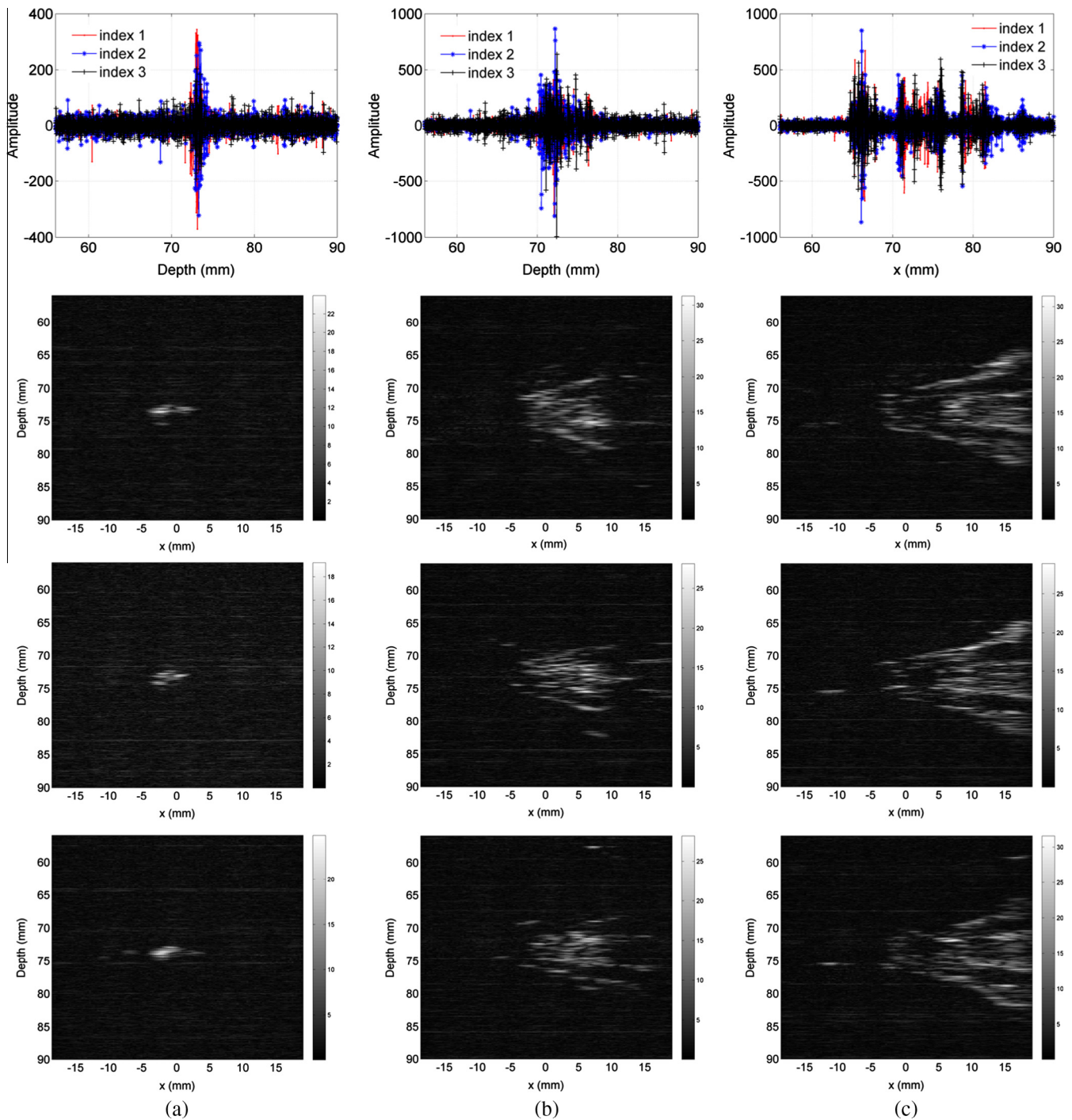


Fig. 3. Results of re-measuring on three different timescales in tap-water at the location of 0 mm. The various T in (a)–(c) were 50 μ s, 5 ms and 20 ms, respectively. D was set to 1.000 ms for the travel time of the sound wave from the HIFU source to the focal region. Zero of the x -axis was set to the center element of the array. In (a) and (b), the first row showed the RF signal of the 64th scan line for both the initial and re-measuring experiments. In (c), the first row showed the RF signal of the 120th scan line for both the initial and re-measuring experiments. The second, third and fourth rows illustrated three times of cavitation maps obtained by ultrasound plane-by-plane scanning method.

algorithms in Matlab. There may be some factors related to the time penalty, including the amount of data, the computer configuration and the beamforming algorithm, etc. The amount of data mainly depends on the number of scan lines per frame, imaging depth and sampling rate. Therefore, this parameter may not be regarded as absolute values of the time consumptions of the different beamforming algorithms. However, it can be used to make the comparisons between different beamforming algorithms. Here, the impacts on image quality and reconstruction time of the post

processing steps, MVCF and MVCS, are demonstrated from both the same simulation and experimental data. All the data processing is performed on a computer with Intel Core i5-2400@3.10 GHz, 64-bit WIN7 system, and 4 GB of RAM.

(1) Simulation

Using the Field II software, we simulated the point targets in different depths to evaluate the beamformers in terms of resolution

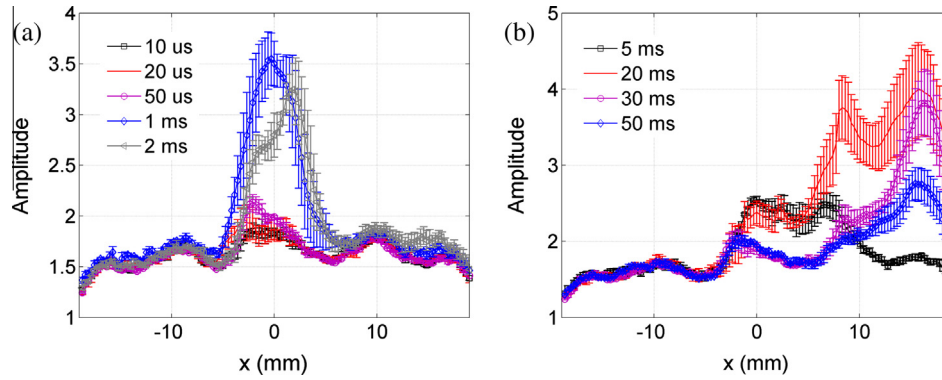


Fig. 4. Gray value curves along the linear array in tap-water at the location of 0 mm. In (a), the various PD were 10 μ s, 20 μ s, 50 μ s, 1 ms and 2 ms, respectively. In (b), the various PD were 5 ms, 20 ms, 30 ms and 50 ms, respectively. D was set to 1.000 ms for the travel time of the sound wave from the HIFU source to the focal region. Zero of the x-axis was set to the center element of the array. The values shown were the average of 20 measures for each T, and the error bars showed the standard deviation.

and SNR, which were determined by the width of the main-lobe and the level of the side-lobe of lateral variation of the point target, respectively. The parameters were set as follows. The linear array transducer was composed of 128 elements with element width of 0.2798 mm, element height of 4 mm, and element kerf of 0.0025 mm. The center frequency of the imaging was 5 MHz, and the sampling frequency was 40 MHz. When performing beamforming, F-num was set as 1, and the maximum active aperture was 64. The imaging region was 38 mm $\times$ 38 mm with line density of 128, and the simulated phantom included 5 pairs of points targets located at depths of 10 mm, 15 mm, 20 mm, 25 mm and 30 mm, respectively, with lateral positions of the two point targets in each pair at the same depth of -3 mm and 3 mm.

Fig. 5 illustrated the comparison among traditional B-mode imaging (Fig. 5(a)) and different beamforming algorithms (Fig. 5 (b)–(d)) in the beamformed imaging based on plane wave using simulated data. Fig. 5(e) and (f) showed the lateral variation of B-mode, DAS, MVCF and MVCS beamformers, at depth (a) 10 mm and (b) 30 mm, respectively. The time penalty for DAS, MVCF and MVCS beamformer based on simulated data, is about 40.91 s, 91.52 s and 30.28 s, respectively.

As shown in Fig. 5(a), the image for B-mode has good resolution and high SNR, corresponding to narrow main-lobe width and low side-lobe level in Fig. 5(e) and (f), due to the line-by-line scan process with focused wave transmission. However, since one frame of B-mode is obtained through sequential scanning, the imaging is a quasi-real-time technique with a limited frame rate. Random distribution and rapid dissolution of cavitation bubbles require high frame rate to acquire the transient state and information, in which B-mode is often unsatisfactory.

In order to image the transient behaviors of cavitation bubbles at one time, a plane wave approach has been developed to monitor and locate the cavitation events. When using plane wave transmission, the imaging frame rate can be enhanced greatly, and beamforming algorithms are needed. The traditional beamforming algorithm is non-adaptive DAS beamforming. As can be seen from Fig. 5(b), (e) and (f), the DAS beamformer has poor resolution and low SNR. The image quality is relatively poor compared to B-mode, and the time penalty is about 40.91 s. Adaptive beamforming calculates the element weight adaptively based on the input signals to reduce the main-lobe and suppress side-lobes. In our work, MVCF is introduced to improve the image quality in terms of resolution and SNR. From the results shown in Fig. 5(c), (e) and (f), the MVCF beamformer provides a significant improvement on resolution and SNR, even better than B-mode image. The point targets can be better resolved. However, the reconstruction time of MVCF is 91.52 s, which is about two times of that of DAS. The CS and

sparse processing method, a novel theory aiming to reduce the amount of data, is introduced to MV beamforming to retain the quality of adaptive beamforming and reduce the calculating data involved. As demonstrated in Fig. 5(d), (e) and (f), the MVCS beamformer results in better resolution but low SNR. However, the time penalty is only 30.28 s, which is significantly less than that of MVCF.

(2) Experiment:

The performances of our algorithms were also studied by detecting the cavitation bubble cloud in water. The duration of HIFU exposure was 5 ms, the center frequency of the imaging was 5 MHz, and the sampling frequency was 40 MHz. When performing beamforming, F-num was set as 1, and the maximum active aperture was 64. The imaging region was 38 mm $\times$ 92 mm with line density of 128. Before beamforming, comb notch filter is applied to filter out the fundamental and harmonic HIFU frequencies to eliminate the interference pulses from the HIFU field.

Fig. 6 showed the images obtained with B-mode imaging and different beamformers displayed over 60 dB dynamic range and the corresponding autocorrelation functions of the enveloped RF data. SNRs (defined as $10\lg(P_{\text{signal}}/P_{\text{noise}})$, where P_{signal} and P_{noise} are the power of signal and noise respectively), are calculated by assuming the signal in the background region (the left rectangular box) as the noise signal, and the signal in the bubble region (the right rectangular box) as the cavitation signal. Moreover, in order to quantify the improvement in resolution due to the beamformers, the autocorrelation functions of the enveloped RF data corresponding to the images are calculated, and resolution gain is defined as the ratio of the -6 dB autocorrelation lengths of the RF data before beamforming to that after beamforming. Table 2 listed the imaging parameters for different beamforming algorithms based on experimental data.

As seen in Fig. 6(b), the DAS beamformer has the resolutions close to those of B-mode (Fig. 6(a)), but also contains high side-lobes. The SNR is 3.70 dB which is higher than 2.58 dB of B-mode, and the time penalty is about 112.14 s. The MVCF beamformer in Fig. 6(c) has an improvement in terms of both resolutions and SNR. As seen in Fig. 6(e) and (f), the lateral and axial resolution of MVCF beamformer has a 2.88 times and 1.88 times higher than those of B-mode. Besides, the SNR of MVCF beamformer has a 14.17 dB higher than that of B-mode. However, the time penalty for MVCF beamformer is about 283.45 s, which is two times of that of DAS beamformer. Compared to DAS beamformer (Fig. 6(b)), the MVCS beamformer (Fig. 6(d)) has a slight increase in resolutions and SNR. Although the image quality of the MVCS beamformer is

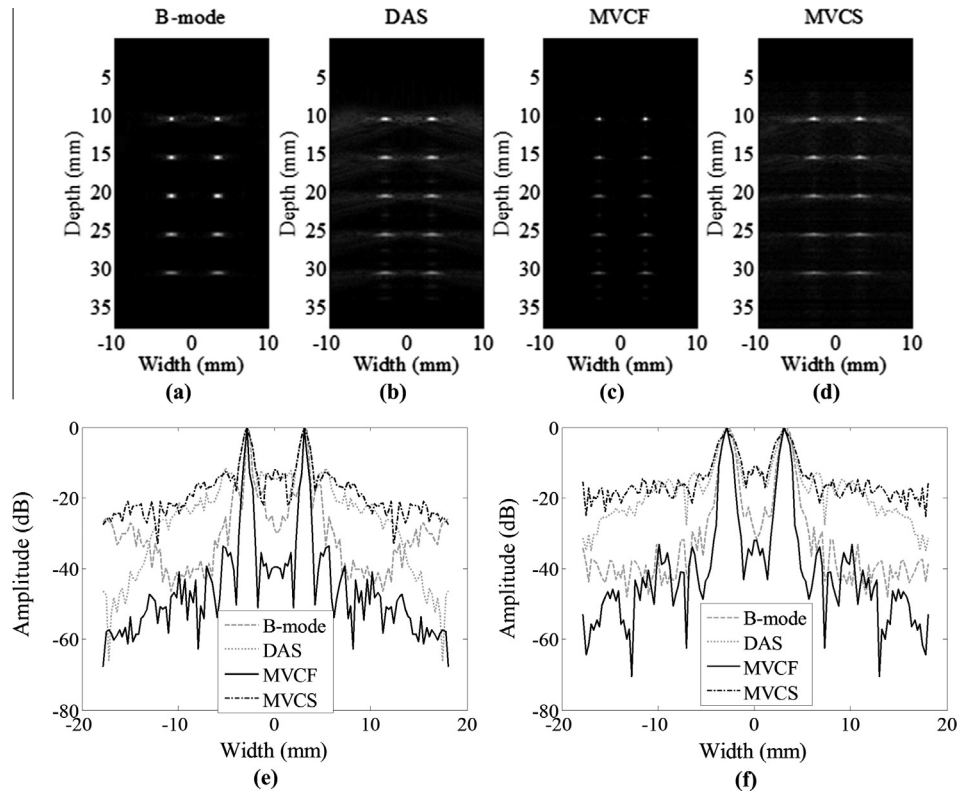


Fig. 5. Comparison among B-mode image and plane wave-based images with different beamforming algorithms of point targets using the simulated data. Images obtained with (a) B-mode, and the (b) DAS, (c) MVCF, and (d) MVCS beamformers displayed with a dynamic range of 60 dB. And lateral variation of B-mode, and DAS, MVCF, and MVCS beamformers, at depth (e) 10 mm and (f) 30 mm.

not as good as that of MVCF beamformer, the time penalty is 42.36 s, which is only 14.94% of that of MVCF beamformer.

After acquiring each output of beamformed RF data, the 3D cavitation map is reconstructed by the Marching Cube algorithm in VTK and the reconstruction time is computed. Here, the 3D reconstruction is performed on an AMD 2.70 GHZ 64-bit computer with 4 GB of RAM. The times consumed on the 3D reconstruction in the phantom vessel with the pulsatile when PD was 20 μ s and in tap-water when PD was 20 ms are about 3.895 ± 0.15 s and 29.766 ± 0.15 s, respectively.

3.3. Temporal evolutions of the cavitation bubble cloud

Fig. 7 showed the temporal evolutions of cavitation bubble cloud in tap-water. The images were obtained by MVCF beamformer for better image quality, and they were provided in an absolute linear scale with the total image depth of 90 mm, and the numbers on the grey scale bar correspond to the strength of echoes. In order to more clearly observe the cavitation bubbles, the depth ranging from 48 mm to 90 mm was displayed here. The temporal resolution of the evolution was set to be several microseconds.

As seen in Fig. 7, the cavitation bubble cloud firstly emerged in the focal region because of the highest acoustic pressure. Later, the area of cavitation bubble cloud symmetrically grew larger in size because inertial cavitation would generate more bubbles that could nucleate additional cavitation activity. With the increase of PD, quasi-continuous streamers of bubbles appeared out of the bulk in the post-focal region, forming a trail of bubbles. Bubbles travelled along directional paths towards junctions with other streamers, which caused a crown-like shape structure. The structure was mainly determined by the distribution of acoustic

pressure. When PD was 5 ms, acoustic pressure in the pre-focal region was large enough to generate cavitation bubbles organized to separate branches. The reason may be that bubbles in the post-focal and focal region could reflect ultrasound energy to the pre-focal region. As PD increased to 30 ms and 50 ms, the streaming became bigger due to acoustic radiation, and the bubbles were pushed farther upward.

The temporal evolution of the cavitation bubble cloud in a phantom vessel with the pulsatile flow is shown in Fig. 8. The images were obtained by MVCF beamformer. The velocity of the flow was 2 cm/s and the pulsation frequency was about 60 times/min. HIFU exposure time ranges from several microseconds to dozens of milliseconds, which was much shorter than pulsation cycle. Therefore, the cavitation field affected by the pulsatile flow immediately after one HIFU exposure can be ignored. After HIFU was turned off for a while, the inertia motion of cavitation bubbles caused by acoustic radiation force (ARF) was reduced by friction and drag force. Furthermore, the pulsatile flow may affect the cavitation field in the vessel. As shown in Fig. 8, when PD was increased from 10 μ s to 50 μ s, the size of the bubble cloud in the vessel became bigger, while the density of the bubble cloud decreased. The bubbles moved in both directions along the vessel, and there was a rip in the bubble cloud when PD was 50 μ s. When PD was increased from 1 ms to 50 ms, the number of cavitation bubbles decreased and became thinner.

3.4. Spatial distribution of cavitation bubble cloud at different locations

Fig. 9 shows spatial distribution of cavitation bubble cloud in tap-water at different locations when PD was 20 ms. The images were obtained by MVCF beamformer. The residual bubbles mainly

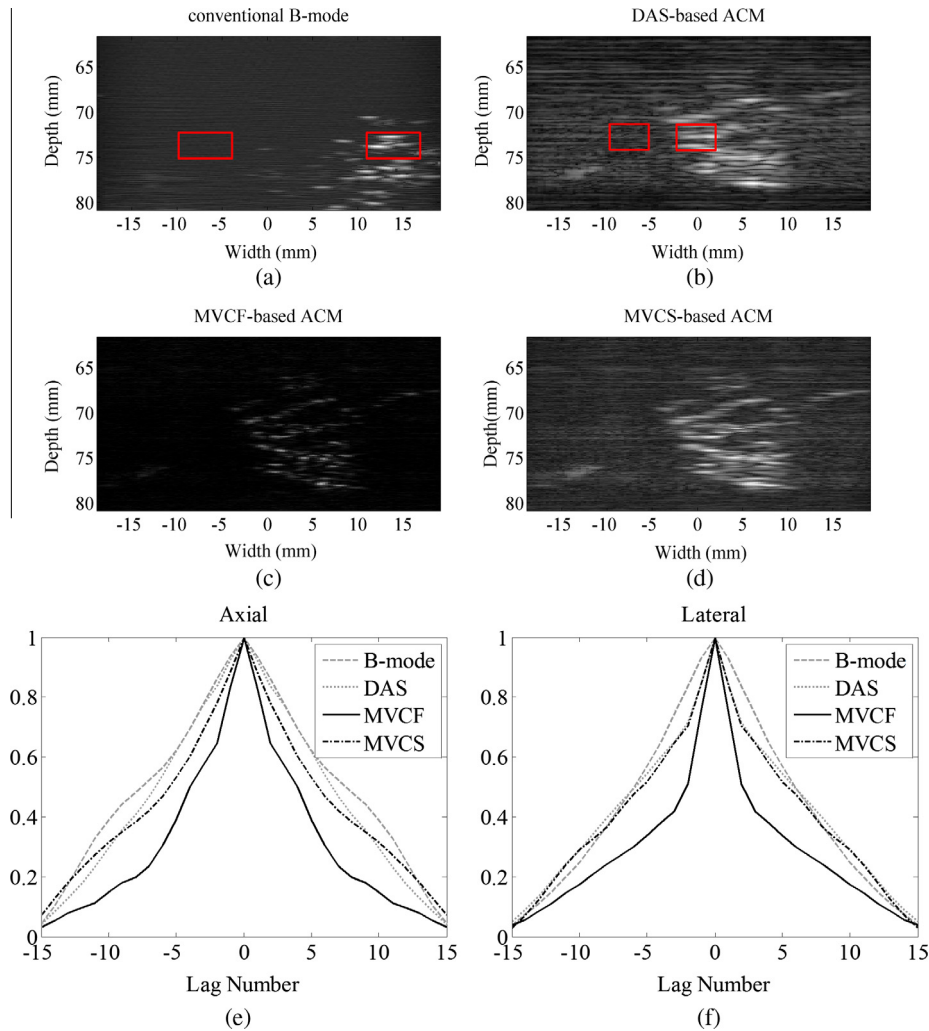


Fig. 6. Comparison among B-mode image and plane wave-based images with different beamforming algorithms of cavitation bubble using the experimental data. Images obtained with (a) B-mode, and the (b) DAS, (c) MVCF, and (d) MVCS beamformers displayed with a dynamic range of 60 dB. And the corresponding autocorrelation functions of the enveloped RF data in axial (e) and lateral direction. The rectangular boxes are used to calculate SNR.

Table 2

Parameters for different beamforming algorithms based on experimental data.

	B	DAS	MVCF	MVCS
Time penalty (s)	/	112.14	283.45	42.36
SNR (dB)	2.58	3.70	16.75	3.95
Axial resolution gain	1	1.16	1.88	1.38
Lateral resolution gain	1	1.02	2.88	1.12

emerged in the focal and post-focal region, forming the structure of streaming bubbles determined by the distribution of acoustic pressure. The spatial distribution of cavitation bubbles at the location of 2 mm showed a distinct cloud structure, and the width along the depth direction was relatively small. By assuming the location of 2 mm as the center, the spatial distribution of cavitation bubbles at different locations moves towards both sides, nearly a symmetry. The width of cavitation bubbles along the depth direction became bigger, while the length of cavitation bubbles along the HIFU axis became smaller with the bubbles focused in the post-focal region. Meanwhile, the tree-crown structure of cavitation bubble became more blurred.

Fig. 10 shows spatial distribution of cavitation bubble cloud in a phantom vessel with the pulsatile flow at different locations when PD was 10 μ s. The images were obtained by MVCF beamformer. The velocity of the flow was 2 cm/s and the pulsation frequency

was about 60 times/min. Since PD was 10 μ s which was much shorter than pulsation cycle, the cavitation field affected by the pulsatile flow immediately after one HIFU exposure can be ignored. The linear array was positioned at each unit location parallel to the axis of HIFU transducer. The spatial distribution of cavitation at the locations of 0 mm showed the bubbles filled in the phantom vessel. Taking the location of 0 mm as a center, the changes of spatial cavitation distribution at different locations on both sides were asymmetric. The pulse-echo signals of cavitation bubbles at the location of 2.5 mm were decreased and nearly invisible, while strong echo region of bubbles at the location of -3 mm could be obviously observed. When further away from the center location, there were smaller size of bubble region. In addition, the distance between the upper and lower vessel walls became smaller due to the cylindrical shape of the vessel. As the linear array had a certain width, the bubble signals at the location of 0 mm could be captured even if the linear array located at the location of -3 mm. Therefore, at the location further away from the center location, bubble region were observed near and upon the upper wall.

3.5. 3D cavitation mapping

Fig. 11 shows 3D cavitation map in tap-water when PD was 20 ms. As seen in Fig. 11, the cavitation activity in 3D space could

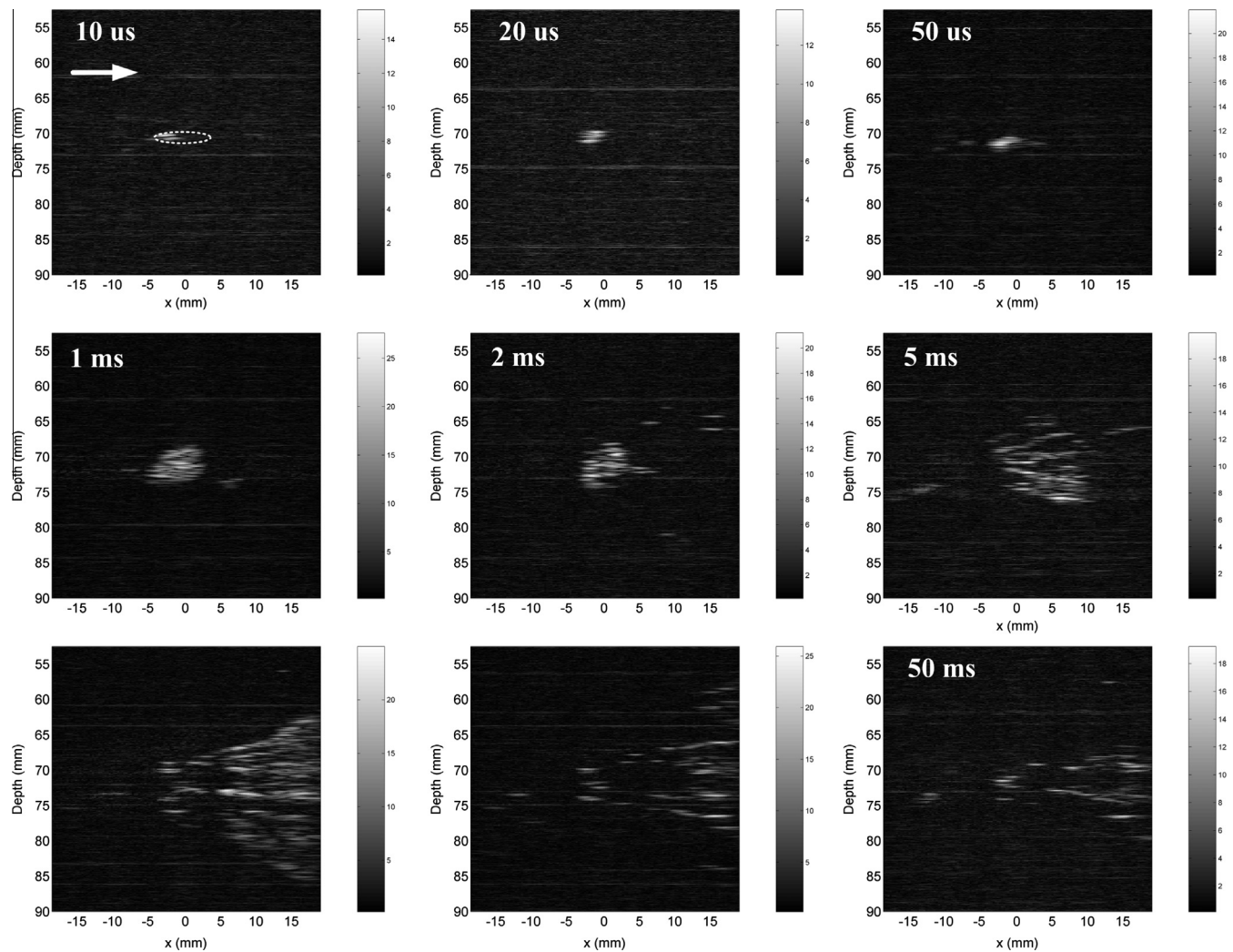


Fig. 7. Temporal evolutions of the cavitation bubble cloud in tap-water at the location of 0 mm. The pointer showed the transmission direction of HIFU, and the dotted line showed the focal region. D was set to 1 ms for the travel time of the sound wave from the HIFU source to the focal region. The various PD were indicated in the left corner of each image. Zero of the x-axis was set to the center element of the array.

be observed. The images of sequence planes were obtained by MVCS beamformer for more rapid data processing. The bubbles mainly appeared in the focal and post-focal region. Besides, there were fewer bubbles in the pre-focal region. This result was consistent with those reported in other published papers, and this 3D cavitation map could provide more complete information about cavitation activity [33,42].

Fig. 12 shows 3D cavitation mapping of a phantom vessel with the pulsatile flow when PD was 20 μ s. The images of sequence planes were obtained by MVCS beamformer for more rapid data processing. The velocity of the flow was 2 cm/s and the pulsation frequency was about 60 times/min. The 3D cavitation map showed the spatial distribution of bubbles in the phantom vessel, from which we could observe that the bubbles located in the middle of the vessel and much closer to the upper wall. The reason may be that the residual bubbles moved upward due to buoyant force.

4. Discussion

The purpose of our study is to develop a 3D-UPACM employing the linear array of a current ultrasound diagnosis system rather than a 2D array to accomplish visualization of 3D cavitation activity. The acquisition of data in 3D is performed by sequential plane-by-plane 2D ultrafast active cavitation mapping. The orientation of

the mechanically scanning is perpendicular to the direction along the linear array. As a consequence, although the step size of 0.5 mm in elevation is different from the element spacing of 0.3 mm, the quality of 3D cavitation maps may not be affected. In addition, in the direction perpendicular to the linear array, though the element of the linear array has a thickness of 0.3 mm, which is smaller than the step size of 0.5 mm, the plane-wave based 2D cavitation map is actually a slice at the center of the element thickness. As a result, the space between two slices is still 0.5 mm. If the resolutions of the 2D cavitation map are better than the step size of 0.5 mm, the space resolutions of the 3D cavitation mapping may be improved by decreasing the step size. Otherwise, further reducing the step size may have no significant improvement in the 3D image quality.

As a stochastic process, the cavitation bubble cloud is unlikely to be perfectly reproduced. However, the results of re-measuring on three different timescales in tap-water (Figs. 3 and 4) showed that three times of cavitation maps appeared very similar and the bubble cloud structure did not change in quality. The reason may be that the bubble cloud is structurally stable on time-scales large compared to the bubble oscillations. Therefore, it may be valid to build up a 3D cavitation map plane-by-plane on the broad scale of the mapping done here. Furthermore, MV adaptive beamforming is combined with CF weighting or CS method for

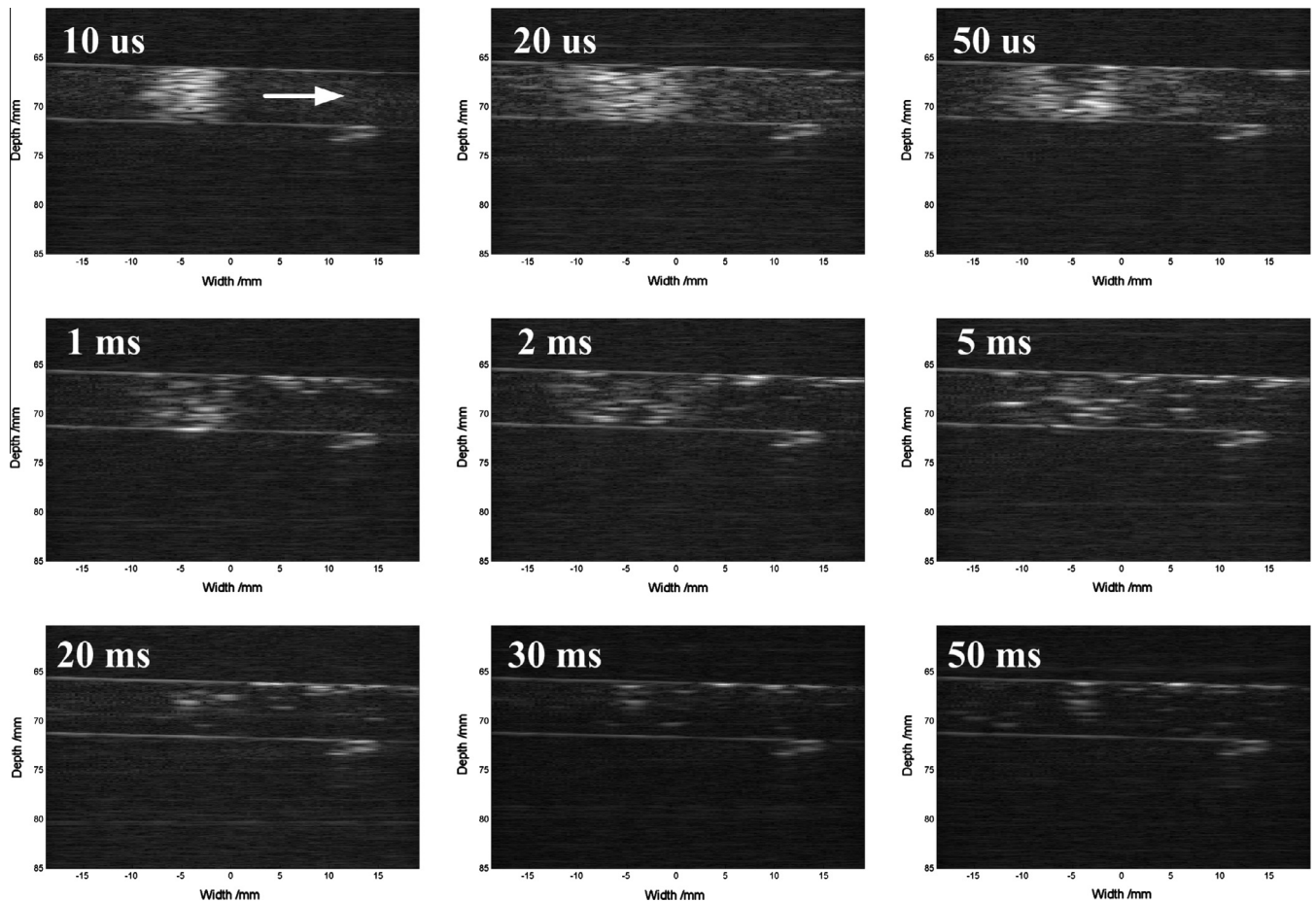


Fig. 8. Temporal evolution of the cavitation bubble cloud in a phantom vessel with the pulsatile flow of 2 cm/s at the location of 0 mm. The pointer showed the flow direction and the transmission direction of HIFU was from inside to outside. The pulsation frequency and the diameter of the vessel were about 60 times/min and 4 mm, respectively. D was set to 1 ms for the travel time of the sound wave from the HIFU source to the focal region. The various PD were indicated in the left corner of each image. Zero of the x -axis was set to the center element of the array.

improved beamforming or more rapid data processing. As shown in Figs. 5 and 6, the MVCF beamformer has a significant improvement in terms of both resolutions and SNR. However, compared with DAS and MVCS beamformer, the MVCF beamformer is a relatively time-consuming process, and the time penalty is about 283.45 s. When imaging quality rather than imaging speed is cared, the MVCF beamformer may be appropriate. The MVCS beamformer can significantly decrease the reconstruction time (only 14.94% time penalty of that of MVCF beamformer), while maintain a slight increase in resolutions and SNR compared to the DAS beamformer. Therefore, this approach may promptly accomplish either 2D or 3D cavitation mapping of the cavitation bubbles with relative high SNR, resolutions and rapid computing. Currently, MVCF beamformer cannot be simply combined with CS theory. The reason is that the CF represents the ratio between the energy of the coherent sum to the total incoherent energy. However, the energy cannot be directly calculated from the beamformed data in the frequency domain of MV. Some revisions about the CF expression are needed, and in our future work, this study will be done, aiming to retain the quality of the MVCF beamformer and also reduce the calculating data involved by CS method.

As shown in Fig. 7, the evolution process of cavitation bubble cloud in tap-water could be divided into three stages: emergence of the cavitation bubble cloud in the focal region (Fig. 7 (PD = 10 μ s, 20 μ s, 50 μ s and 1 ms)), formation of the bubble cloud in the post-focal region and pre-focal region (Fig. 7 (PD = 2 ms, 5 ms and 20 ms)), and the reduction of the bubble cloud (Fig. 7 (PD = 30 ms

and 50 ms)). Cavitation bubble cloud firstly emerged in the focal region because of the highest acoustic pressure. Later, due to the primary Bjerknes force, the bubbles continuously generated in the focal region may move out the focal region along specific paths in the post-focal region thus forming streamers with a crown-like shape. As driven by the acoustic field, the bubbles can undergo pinch-off or even explosions. Then, sub-micron “daughter” bubbles may be produced, which may serve as fresh cavitation nuclei. Probably, these bubbles migrate to the apparent bubble source and merge to visible size, feeding the stream of bubbles. Also, bubbles are formed in the pre-focal region, which has the secondary maximum acoustic intensity, especially the region close to focal region. Moreover, with fresh nuclei into the focus, bubbles concentrated in the focal region may reflect the acoustic wave, and strengthen the pressure in the pre-focal region, which allowed more bubbles to be generated. As more bubbles are generated in the pre-focal region, the small oscillating bubbles may approach each other, and coalesce to form a larger single bubble due to the secondary Bjerknes forces. As PD increased, the bubbles in the pre-focal and focal region may increase the attenuation of acoustic power, and thus bubbles in the post-focal region were insufficiently generated. As shown in Fig. 8 (PD = 50 μ s), there was a rip in the bubble cloud. Two possible reasons are as follows. One is the slow-time steady flow pattern created by acoustic streaming. The fluid accelerates in the focus of HIFU, and the vortices are created laterally to either side of the focus since the presence of the cylindrical boundary. These arise due to the necessity of a fluid return path to satisfy

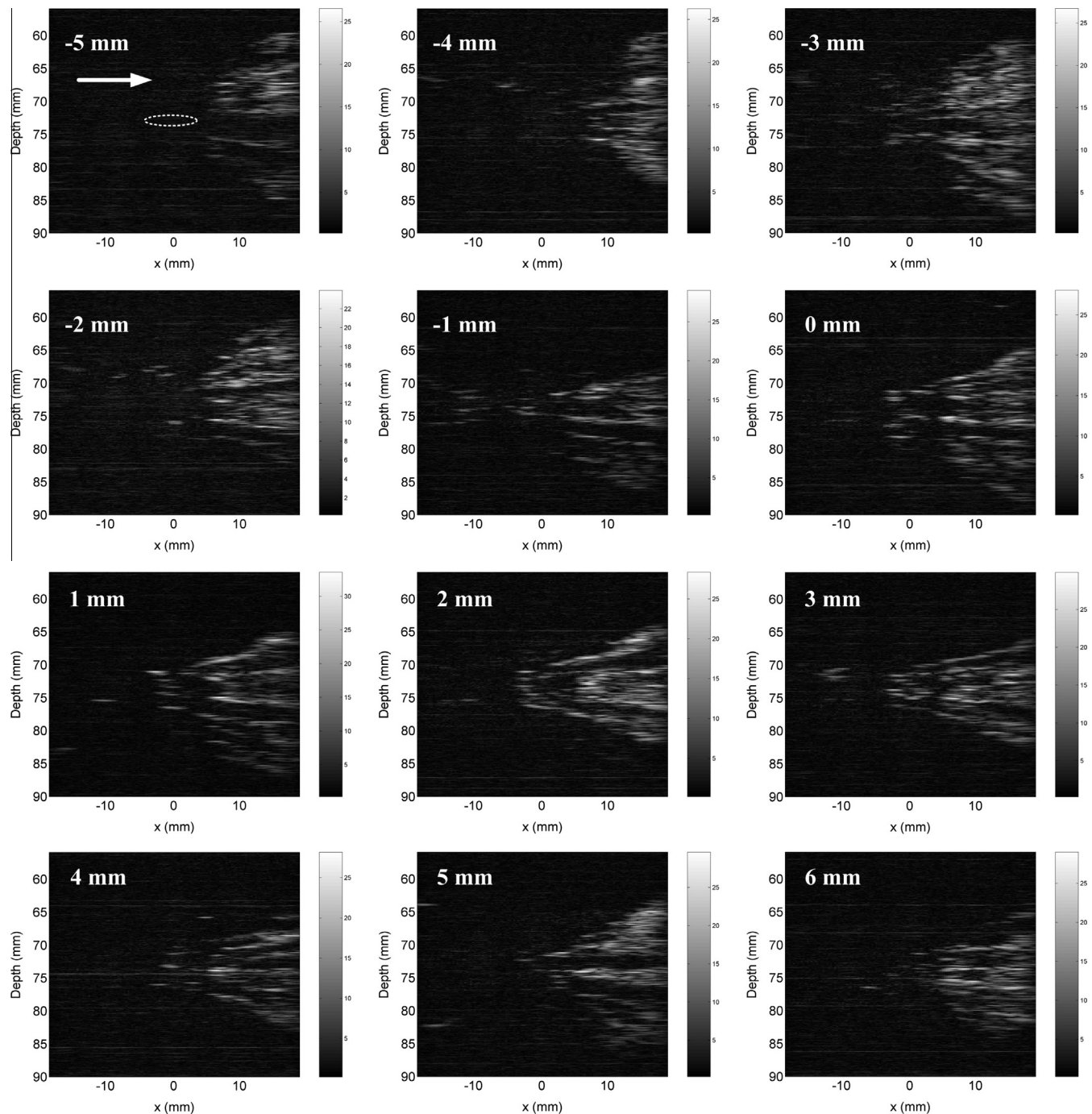


Fig. 9. The spatial distribution of cavitation bubble cloud in tap-water at different locations when PD was 20 ms. The pointer showed the transmission direction of HIFU, and the dotted line showed the focal region. D was set to 1 ms for the travel time of the sound wave from the HIFU source to the focal region. The various locations were indicated in the left corner of each image. Zero of the x -axis was set to the center element of the array.

continuity of the fluid volume. Additionally, the fluid flow velocity increases as PD increases. The other one is the motions of cavitation bubble cloud caused by ARF. When PD is increased to 50 μ s, the motion velocities caused by ARF increase. Also, due to the presence of the cylindrical boundary, the bubbles take the focus as the center and move in both directions along the vessel. As a result, a rip is formed in the bubble cloud. Since the HIFU pulse duration and the delay between the end of the sonication and the imaging pulse which is carried out with a plane wave approach can be finely tuned, the time interval between temporal evolutions of cavitation bubble cloud could be several microseconds. Therefore, 3D

visualization of the time evolution of bubble activity can be achieved.

This 3D-UPACM technique requires a gassy fluid medium or a constant supply of new cavitation nuclei in order to image at each slice. And the limitation may render the 3D cavitation mapping difficult if not impossible to be used in clinics. Besides, the 3D-UPACM is performed after HIFU exposure, and thus cannot provide the bubble activity during the sonication as well as the information about the type of oscillation. However, this 3D-UPACM technique can be used to visualize residual bubbles after HIFU exposure. In the histotripsy process, residual bubbles in the intervals of HIFU

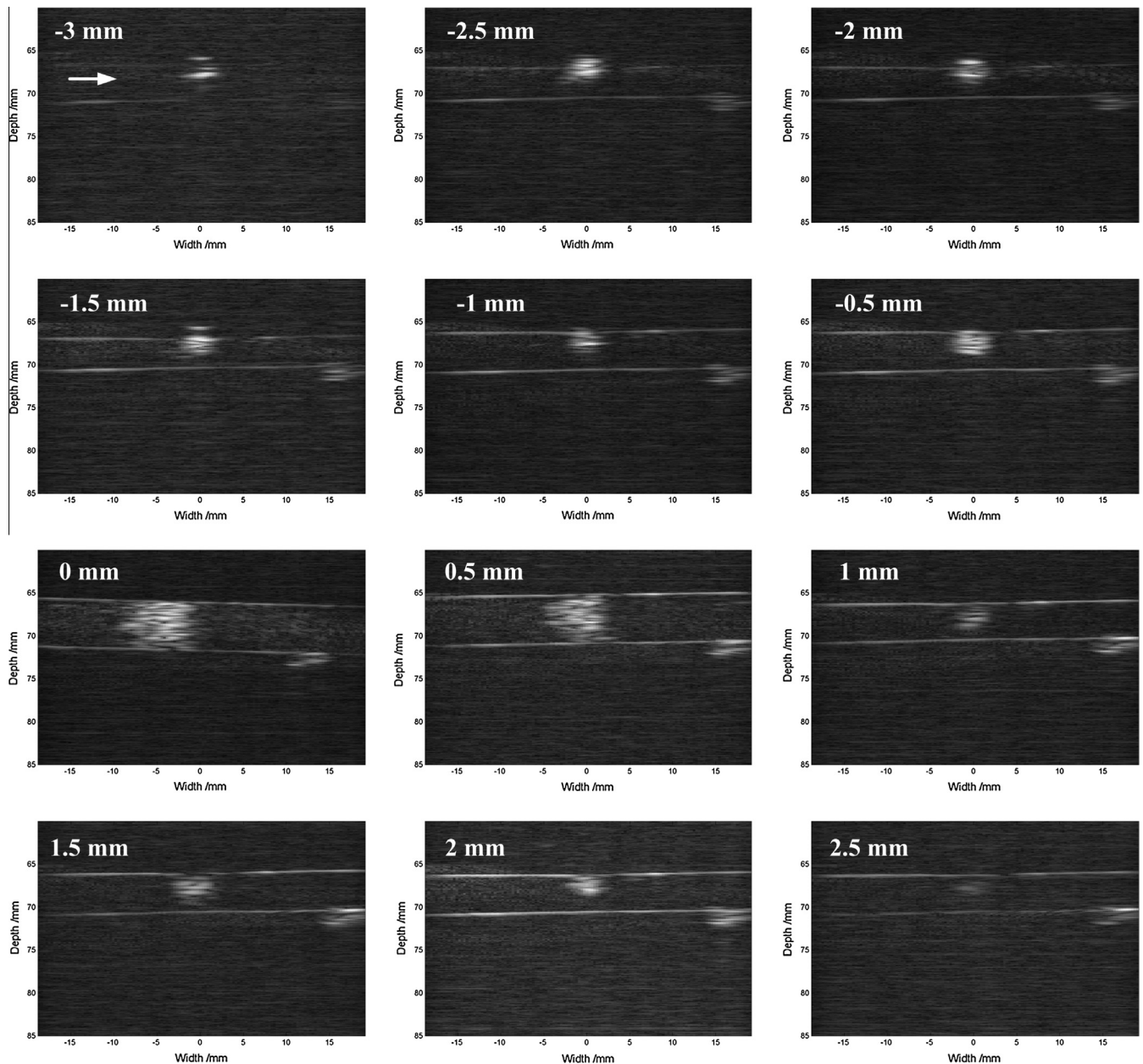


Fig. 10. The spatial distribution of cavitation bubble cloud in a phantom vessel with the pulsatile flow of 2 cm/s at different locations when PD was 10 μ s. The pointer showed the flow direction and the transmission direction of HIFU was from inside to outside. The pulsation frequency and the diameter of the vessel were about 60 times/min and 4 mm, respectively. D was set to 1 ms for the travel time of the sound wave from the HIFU source to the focal region. The various locations were indicated in the left corner of each image. Zero of the x-axis was set to the center element of the array.

exposures can survive for ten of millisecond [10], and have a dual impact on the histotripsy pulses. On the one hand, they may serve as nuclei for easier generation of cavitation events, but on the other hand they act as strong scattering particles in the sound propagation path causing energy “shadowing”. The initiation and maintenance of cavitation bubbles are vital in the treatment. Successive, intense, and very short pulse ($<50 \mu$ s) delivered at low duty cycles ($<5\%$) can obtain high efficiency [10,52,53]. For example, pulse during and duty cycle is set to 20μ s and 5% respectively, which means the HIFU off-time is 380μ s. During such short HIFU off-time, conventional ultrasound imaging with a limited frame rate is not suitable for monitoring, in this case, due to serious interference in the scan lines by the HIFU field. As the time interval between temporal evolutions could be several microseconds, this 3D-UPACM technique can be employed to investigate the 3D

spatial-temporal evolution of residual bubbles. Besides, the dissolving process of corresponding bubbles can be acquired through varying the parameters D in Fig. 1(b). Moreover, this technique may be useful in a laboratory environment where more careful analysis may be required under controlled conditions. Although these investigation did not directly aim at the clinical applications, the 3D cavitation maps obtained may provide a reference for clinical HIFU therapy.

5. Conclusion

In this study, a 3D-UPACM based on the linear array of a current ultrasound diagnosis system was proposed. The feasibility of this technique was demonstrated in tap-water and a phantom vessel with the pulsatile flow. We observed temporal evolutions and

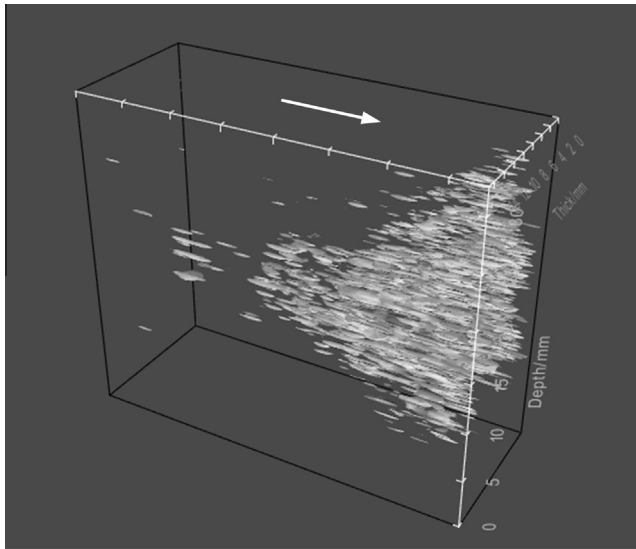


Fig. 11. 3D cavitation map in tap-water when PD was 20 ms. The pointer showed the transmission direction of HIFU. D was set to 1 ms for the travel time of the sound wave from the HIFU source to the focal region.

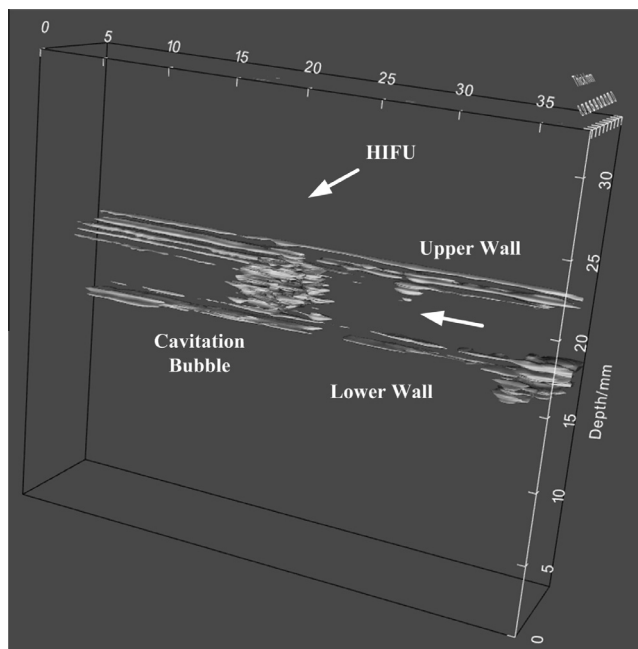


Fig. 12. 3D cavitation mapping in a phantom vessel with the pulsatile flow of 2 cm/s when PD was 20 μ s. The pointer in the vessel showed the flow direction, while the pointer on the vessel showed the ultrasound propagation direction. The pulsation frequency and the diameter of the vessel were about 60 times/min and 4 mm, respectively. D was set to 1 ms for the travel time of the sound wave from the HIFU source to the focal region.

spatial distributions of cavitation bubble cloud. The time interval between temporal evolutions of cavitation bubble cloud could be several microseconds, and through hardware control it could even reach a sub-microsecond level. MV adaptive beamforming is combined with CF weighting or CS method to process the raw RF data, resulting in cavitation maps with high image quality or high reconstruction speed. MVCF beamformer had a signal-to-noise ratio (SNR) at 14.17 dB higher, lateral and axial resolution at 2.88 times and 1.88 times, compared to those of B-mode. MVCS beamformer had only 14.94% time penalty of that of MVCF

beamformer. Additionally, as only a plane wave transmission were performed after one HIFU exposure, the interactions between active detection and bubble behavior were reduced. With this technique, 3D cavitation map containing more information of cavitation activity could be obtained by using the linear array of a current ultrasound diagnosis system rather than a 2D array transducer, therefore decreasing the cost of the instrument. Although the application is limited by the requirement for a gassy fluid that allows replenishment of nuclei between HIFU exposures, this 3D-UPACM may exhibit a useful tool in 3D cavitation mapping for HIFU with high speed, precision and resolution, especially in a laboratory environment where more careful analysis may be required under controlled conditions.

Acknowledgments

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